A Blockchain-based Identity Authentication Protocol for Edge IoT Devices Management

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***Abstract*—With the widespread application of edge IoT devices, the Internet of everything has become a reality. It has profoundly affected this digital economy era and brought great convenience to people's lives. However, security incidents involving the leakage of user-sensitive information due to uncontrolled access rights in edge IoT systems have occurred frequently in recent years. Besides, users need to use different authentication methods when using edge devices from different vendors, which also increases the user's burden. These problems bring great threats and challenges to the development of the IoT ecosystem. Therefore, how to design a more sustainable and smart digital identity authentication mechanism for edge IoT devices management is still a problem worthy of research. In this paper, we have proposed and implemented a blockchain-based identity authentication protocol for edge IoT devices to deal with the security threats of the existing centralized identity authentication model and solve the problem of low authentication efficiency of existing heterogeneous devices, which makes the management of edge devices more intelligent. Our experiments and performance analysis show the merits of our proposed protocol in terms of security and efficiency.**

I. INTRODUCTION

There are tens of billions of edge devices connected to the IoT in the world at present. Cyberspace intelligent systems built by edge IoT devices have brought great benefits to human life. A major feature of the IoT world is that there are a large number of edge devices that need to perform specific intelligent procedures based on the authentication of trusted identities. However, the identity authentication of edge devices seems to have always been the target of attacks in recent years. The hacking of massive edge IoT devices with a single password authentication method has caused huge damage to user privacy, such as the famous Amazon's Ring camera incident. It is also a big challenge to design a unified IoT authentication interface to deal with IoT devices of a wide variety of vendors. Many users choose to use multiple edge IoT devices from different vendors due to the function and cost performance and other factors. As a result, users are required to conduct different authentication procedures each time they use the devices, which undoubtedly increases the burden on users and affects their experience because existing edge IoT devices management does not reach the so-called smart level claimed by vendors. Moreover, this issue also greatly hinders the development of the current IoT ecosystem, and the data of separated IoT devices cannot be shared to achieve the value of the data itself. Therefore, in the current IoT development process, designing a more sustainable and intelligent identity authentication mechanism for edge devices management is still a problem worthy of in-depth study.

There have been some researches on edge devices identity authentication before. In

2016, Neto et al. proposed identity authentication based on identity control and attribute-based encryption to implement the IoT system [1]. In [2], the authors proposed an end-to-end authentication key exchange protocol for wearable devices in the smart environment and optimized the key authentication process in [3]. [4] proposed a zero-knowledge proof-based identity verification and key exchange protocol, and used the new cryptographic techniques to innovate the original edge devices authentication. [5] combined the idea of identity-based encryption (IBE), physically unclonable functions (PUFs), and the key hash function to develop a device authentication and key exchange protocol. In [6], the authors realized the identity authentication system of IoT devices by reading the EEG (electroencephalography) of edge devices. M. Miettinen et al. extracted relevant features from the traffic packets of edge devices to implement device identity authentication and combined the classifier method to identify vulnerable devices [7]. The researchers used unique addressing and identification methods to authenticate smart medical monitoring devices so that they can be uniquely identified in the network of the medical devices in [8], but the authentication process still relies on a centralized server.

Although the above methods implemented the identity authentication of edge devices based on cryptography and hardware, many of these tasks stay at the level of the system model, and the corresponding operations require high costs, which are not available in the current large-scale edge devices network. It is necessary to explore and design a more secure, efficient, and sustainable IoT identity authentication

mechanism.

Blockchain technology is a creative application of distributed data storage, point-to-point transmission, consensus mechanism, encryption algorithm, and other technologies. The distributed architecture of the blockchain is very suitable for providing edge devices identification, identity verification, and data seamless connection and transparent transmission. The realization of edge IoT devices identity authentication based on blockchain technology has become one of the current research hotspots. Recently, some well-known IT service providers such as Microsoft, IBM, AWS, etc. have opened up services that combine blockchain and edge IoT nodes, and more and more companies have adopted blockchain technology to develop the latest identity authentication technology for edge IoT devices.

This paper proposes a blockchain-based identity authentication protocol for edge devices to cope with the security threats of the existing centralized identity authentication model and solve the problem of low authentication efficiency of existing heterogeneous edge devices, and achieve more sustainable and efficient digital identity authentication of IoT devices. Particularly, the main contributions of our work are as follows.

- We propose a blockchain-based edge IoT devices identity authentication protocol. Our solution extracts multi-dimensional characteristics of edge devices and combines encryption and signatures to generate IDs for edge devices and stores them in the blockchain. The users only need to verify the public ledger

information of the blockchain to finish the identity authentication of the edge devices.

- We propose a new unified authentication method for edge devices users' identities authentication. We bind the users' and the edge devices' identities through the two-factor authentication method of users' fingerprints and mobile phone text messages, which achieves more intelligent and secure multi-device authentication from different vendors.
- We implement the proposed protocol and analyze its performance.

The rest of this paper is arranged as follows: In section II, we review some related works on identity authentication. In section III, we introduce some background concepts. We describe our overall architecture of the proposed protocol in section IV. Section V gives the reference implementation of the proposed protocol. We perform some experiments and performance analyses of our solution in section VI. Finally, we conclude this paper in section VII.

II. RELATED WORK

Many researchers have done a lot of research on the identity authentication of edge IoT devices. In [9], Z. Wang et al. developed a privacy protection authentication protocol for edge devices, which integrated a modified short group signature structure and Shamir's secret sharing scheme, but its scalability needs to be further explored. [10] proposed an authentication framework that provides fine-grained protection for

embedded devices. The framework is based on a series of PUF circuits implanted in each chip of the device, requiring special hardware settings. Recently, there has been some work in applying blockchain to edge devices identity authentication. In [11], a consortium blockchain platform is used for identity authentication of edge devices to prevent authenticated devices from failed authentication due to attacks on the central server. In [12], Ronghua et al. propose a blockchain-based system that aims to perform an effective access control process for devices in large-scale IoT systems with smart contracts. [13] proposed an edge devices authentication scheme based on a hybrid blockchain including a local blockchain and a public blockchain. In [14], the author designed a blockchain connection gateway based on the Ethereum blockchain platform and smart contracts to help users better protect their privacy when using IoT devices. In [15]-[18], researchers extracted multi-dimensional features by analyzing network traffic to strengthen the protection of users' privacy. In [19], Rahman et al. combined SDN and blockchain to set relevant rules to deal with security attacks in IoT device authentication. Also, in some studies such as [20]-[25], the authors apply blockchain technology to edge device networks, 5G base station authentication, wireless sensor networks, fog computing networks, and urban smart infrastructure devices management. These solutions provide many architectural references for exploring the application of distributed ledger technology to the identity authentication of edge IoT devices. It is worth noting that there are some research and application of economic incentives and consensus mechanisms such as reward and punishment measures to maintain the ecosystem of IoT devices and maintain the

trusted environment of devices networks. [26]-[29] proposed the identity authentication schemes based on reputation mechanisms, thereby building the credible environment of edge nodes.

III. PRELIMINARIES

A. Blockchain

Blockchain is essentially a shared database, and the data or information stored in it is jointly maintained by all nodes. In the blockchain, all network participants have the right to access the distributed ledger and its unalterable transaction records. When using this distributed ledger, each transaction is recorded only once, thereby eliminating the typical duplication of work in traditional business networks. When each transaction occurs, it will be recorded as a data block. The data block can record any information, such as the factory information of the IoT device. Moreover, each data block in the blockchain is connected to the blocks before and after it, and these blocks constitute a data chain. Blocks can confirm the exact time and sequence of transactions, and by linking the blocks together, it can prevent any block from being changed or inserted between two existing blocks. Each additional block will strengthen the verification of the previous block, thereby enhancing the entire blockchain. This enables the blockchain to prevent tampering and provides key advantages that cannot be changed. This eliminates the possibility of tampering by malicious participants and builds a transaction ledger that you and other blockchain members can trust. After the transaction is recorded in the blockchain ledger, no

participant can change or tamper with the transaction. According to the degree of openness of the blockchain, it can be divided into three types: public blockchain, consortium blockchain, and private blockchain.

B. Hash algorithm

The hash algorithm is a one-way cryptosystem, which is an irreversible mapping from plaintext to ciphertext. There is only an encryption process and no decryption process. The purpose of the hash algorithm is to verify whether the original data has been tampered with. An n-bit hash function is a mapping from an arbitrarily long message to an n-bit hash value, and an n-bit cryptographic hash function is a one-way, collision-avoided n-bit hash function. Such functions are currently extremely important means in digital signature and password protection. Currently, the more popular hash functions mainly include MD5, SHA-1, SHA-2 series, etc.

IV. OVERVIEW OF THE WORKFLOW

The shortcomings of existing edge devices authentication can be overcome by the openness, transparency, and non-tamperability of blockchain technology. Therefore, we propose a blockchain-based edge IoT devices identity authentication protocol to achieve a more sustainable and efficient digital identity authentication. This section mainly introduces the structure and main workflow of the proposed protocol.

A. Definition of related roles

IoT vendors: They are responsible for the production of IoT devices and generating corresponding IDs for IoT devices for subsequent certification.Vendors are the

goalkeepers of devices security authentication, and they don't want their products to be attacked, which will affect their reputation and economic interests.

Users: As users of IoT devices, they upload their data to devices to improve the convenience of life. Meantime, the users' data involves a large amount of personal privacy, such as the users' location, private photos, etc., once it is leaked, it will pose a great threat to personal privacy.

IoT unified authentication platform: This platform acts as a gateway. It mainly helps IoT devices(Users) and vendors communicate with the blockchain network. It is also responsible for the management of the IoT blockchain network such as traceability audits, node verification request processing, etc. Generally, this platform is operated by an official organization or a standards association recognized by the IoT industry, because in the future there will be more than tens of billions of edge devices, and there are certain differences and conflicts of interest among different brands and different vendors. Besides, the users also need to interact with the underlying IoT blockchain network through a good interface. Therefore, building a unified IoT blockchain authentication platform and setting up corresponding administrators is undoubtedly necessary for the entire IoT devices ecosystem.

B. The architecture description

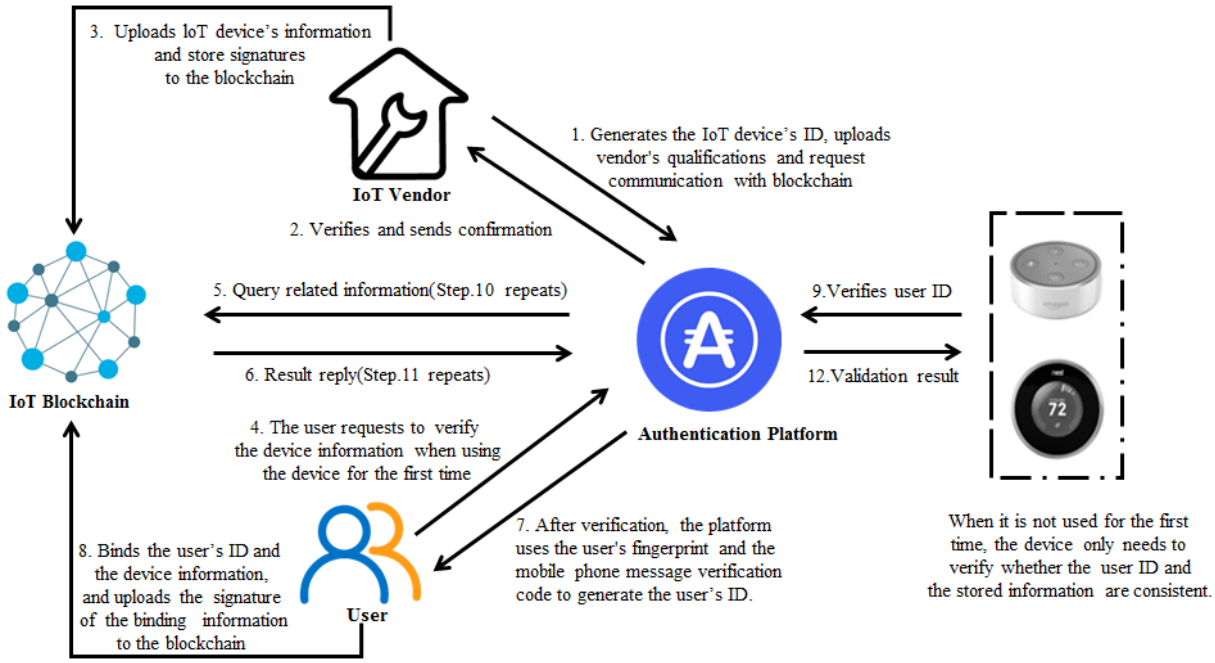

Fig. 1. The architecture of the proposed protocol

Fig. 1. defines the architecture of our protocol, which describes the workflow of our proposed protocol in detail. The following is a general overview of these processes:

Step 1: Vendors need to generate IDs for IoT devices in the production phase. The specific method is to obtain the IoT device's multi-dimensional features information and calculate and generate a unique identification ID that can represent the current status of the IoT device. Multi-dimensional features include the device's hardware information, operating system information, serial number, vendor information, etc.

Step 2: After the ID is generated for the IoT device, the vendor begins to register the IoT device's information in the IoT blockchain network. The IoT vendor needs to send the relevant qualifications to the IoT unified authentication platform to verify the vendor's identity, and then sends the IoT device information registration request.

Step 3: After the authentication platform confirms the request, the IoT vendor will create the hash of the IoT device's ID, the features information of the IoT device

(such as hardware information, operation system information, serial number, vendor information, etc.). Then it is signed with the vendor's private key and published to the public ledger of the blockchain network to register the ID of the IoT device in the blockchain.

Step 4: The user needs to verify whether the edge device is a credible device through the IoT blockchain when using the IoT device for the first time. Specifically, when the device is started for the first time, the user will first check the ID of the device, the corresponding device information, and related signature. And the user will send the device verification request to the blockchain through the platform.

Step 5: The IoT authentication platform indexes the device's information stored in the Blockchain by the new device's information released by the user, and if there is relevant information then the platform compares whether the two contents are consistent.

Step 6: After the IoT blockchain receives the verification request, the blockchain nodes need to verify whether the information is a legitimate device. Once the verification is passed, the IoT device can access the blockchain network.

Step 7: Once the IoT device identification procedure is completed, the authentication platform combines the user's fingerprint information and mobile phone verification code to generate the user's ID.

Step 8: The user binds the IoT device information with the identity of the device owner, that is, associates the IoT device ID with the user's ID to achieve device authentication, and the user can also set the system usage preferences of the device

and other information. Then the user creates a signature based on the device ID, the features information of the device, the user's ID, the vendor's signature information, device usage preferences, and other information and released it to the blockchain network. In this way, the user can use the device after completing the first authentication.

Step 9-12: When the device is not used for the first time, the IoT authentication platform only needs to compare the user’s ID and the device ID to verify the user's identity. That is, after the user enters the ID, the authentication platform queries whether the device ID bound to the corresponding user’s ID is consistent with the one own by the user. If the verification result is consistent, it means that the device belongs to the user, and the user authentication is completed. Through this solution, different types of IoT devices can be associated with users’ IDs in the blockchain network to achieve interconnection between different types of edge devices. Users can also achieve business requirements such as authorized access and information sharing with different IoT devices through the blockchain.

V. IMPLEMENTATION OF THE PROPOSED PROTOCOL

This section mainly introduces the communication process through the symbolic description and is a reference implementation for the proposed protocol. Table 1 gives the definitions and corresponding meanings involved in the proposed protocol.

TABLE I

UNITS FOR MAGNETIC PROPERTIES

Notation	Description
Vendor	The IoT vendor and related information
PK_V	The IoT vendor's public key
SK_V	The IoT vendor's private key
Hash	Hash function
H_n	Hash function for n
Signature	Signature function
Sig_m	Signature for m
OS	Operating system information of device
SN	Serial number of the IoT device
Device	Hardware information of IoT device
ID_{device}	The ID of IoT device
PK_U	The user's public key
SK_U	The user's private key
ID_{user}	The User's ID
Fingerprint	The User's fingerprint information
$Rquest_{A->B}$	A sends a request to B
$Reply_{A->B}$	A sends a reply to B
ACK	Acknowledgment number
IoTA	IoT Authentication Platform
IoTB	IoT Blockchain
Tx	Send a transaction
Txid	Transaction ID

A. Device information registration phase

The specific communication process in this phase is shown in Fig. 2.

(1) *Vendor* : Generates ID_{device}. IoT vendor uses the following method to generate an ID for an IoT device:

$$ID_{device}=H(H(SN)+H(OS)+H(Vendor)+H(Device)) \quad (1)$$

(2) *Vendor* → *IoTA:* {*Vendor*}

(3) *IoTA* → *Vendor*: {$Reply_{IoTA->vendor}$}, sends the confirmation result.

(4) *Vendor* → *IoTA* → *IoTB:* { $Rquest_{vendor->IoTB}$ },sends a registration request.

(5) *IoTB* → *Vendor*: { $Reply_{IoTB->vendor}$ },establishes a communication channel.

(6) *Vendor* →*IoTB*: Sends a *Tx*{ Sig_{device}}

$$H_{device}=H(H(SN)+H(OS)+H(Vendors)+H(Device)+H(ID_{device})) \quad (2)$$

$$Sig_{device}=Signature(H_{device},SK_V) \quad (3)$$

(7) *IoTB* → *Vendor*: {*Txid*} (it can be used for query of IoT device information)

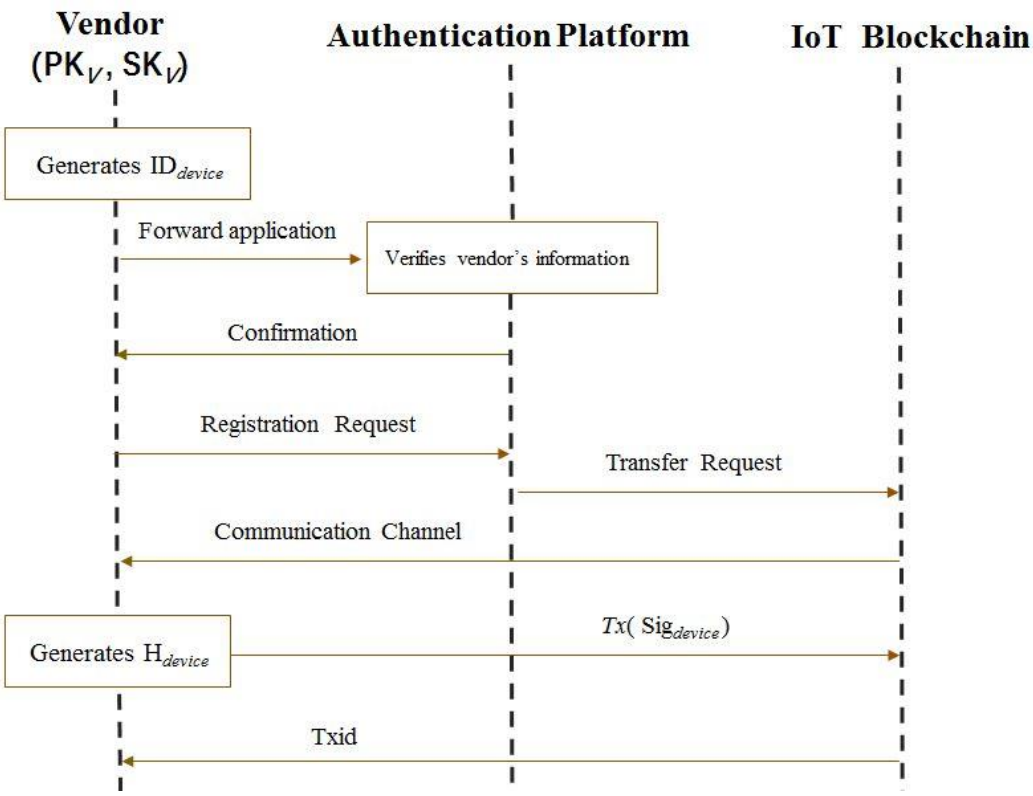

Fig. 2. Device information registration phase

B. Device authentication phase

(1) *User*(*Device*): Starts the IoT device for the first time, and gets {ID_{device}}

(2) *User*(*Device*) →*IoTA*→ *IoTB:* {ID_{device}, $Rquest_{user->IoTB}$}

(3) *IoTA*→ *User*(*Device*)*:* {*ACK*}, if the data format is correct, it will be forwarded

(4) *IoTB*→*IoTA*→*User*(*Device*): {*Txid*}(if exists) *,and user gets the original* ID'_{device}

(5) *IoTB*: Verifies the original ID'_{device} and the ID_{device} uploaded by user,

If (ID'_{device} == ID_{device}) and accepts the new device.

(6) *IoTB* →*IoTA*→ *Device*: { $Reply_{IoTB->device}$ },sends the verification result

(7) *Device*→ *IoTB*: Establishes a connection (if the verification is successful) and joins the IoT Blockchain network.

(8) Finish the IoT device authentication phase.

The above negotiation process in this phase is shown in Fig. 3.

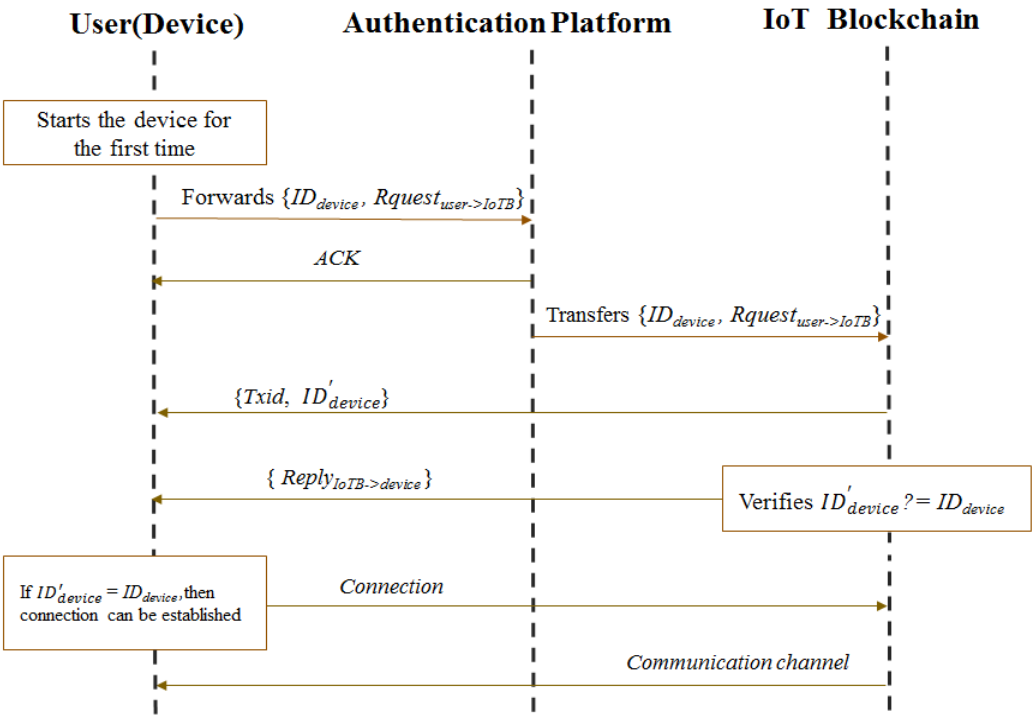

Fig. 3. Device authentication phase

C. User Authentication phase

The negotiation process in the user authentication phase is shown in Fig. 4. Through the communication in this process, the fast authentication between the user and the device can be realized, which is helpful to improve the intelligence of edge devices.

(1) *User(Device)* →*IoTA* →*IoTB*: Completed the phase of IoT devices authentication.

(2) *User* →*IoTA* : Sends message Verification request

(3) *IoTA* → *User* : Message code {*M*}

(4) *User*: Generates ID_{user} as the following method.

$$ID_{user}=H(H(Fingerprint)+H(M)) \quad (4)$$

(5) *User*: Binds {ID_{user}:ID_{device}}, set device usage preferences and related information .

(6) *User* → *IoTA* → *IoTB*: Sends a *Tx*{Sig_{user}}

$$Sig_{user} = Signature(\{ ID_{user}, ID_{device}, H_{device}, \text{preferences}, Sig_{device} \}, SK_U) \quad (5)$$

(7) *User* → *IoTA* → *IoTB*: Requests to use the IoT device again.

(8) *IoTB*: Verifies {ID_{user} :ID_{device} }.

(9) *IoTB* → *IoTA* → *User*: { $Reply_{IoTB->user}$ }, sends the verification result

(10) If ID_{user} matches ID_{device}, then finishes this phase.

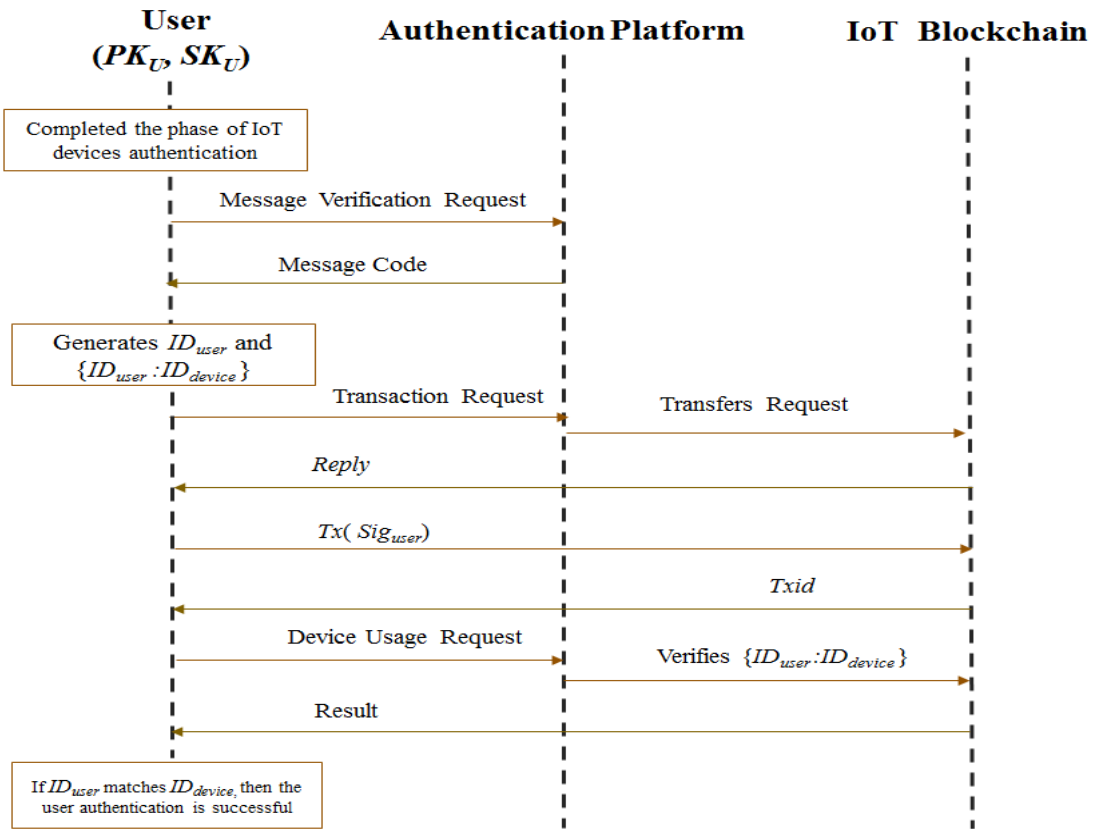

Fig. 4. The user authentication phase

VI. EXPERIMENTAL DESIGN AND ANALYSIS

A. Experiment Setup

The dataset in this experiment is collected from the laboratory because the features of the edge devices are very accessible online. We collected 30 smart edge devices of different vendors, including sweeping robots (10), smart door locks (10), and smart cameras (10). The devices in each type come from 5 different vendors, each vendor with 2 devices. The experiment extracted several features from these devices, including serial number (SN), vendor information(vendor), operating system information (OS), hardware information (device), production date, and other valuable information, and we standardized the collected data and unified it into the same structured data for subsequent processing. The data size of the extracting device features and the time distribution of their manufacture are presented in Fig. 5.

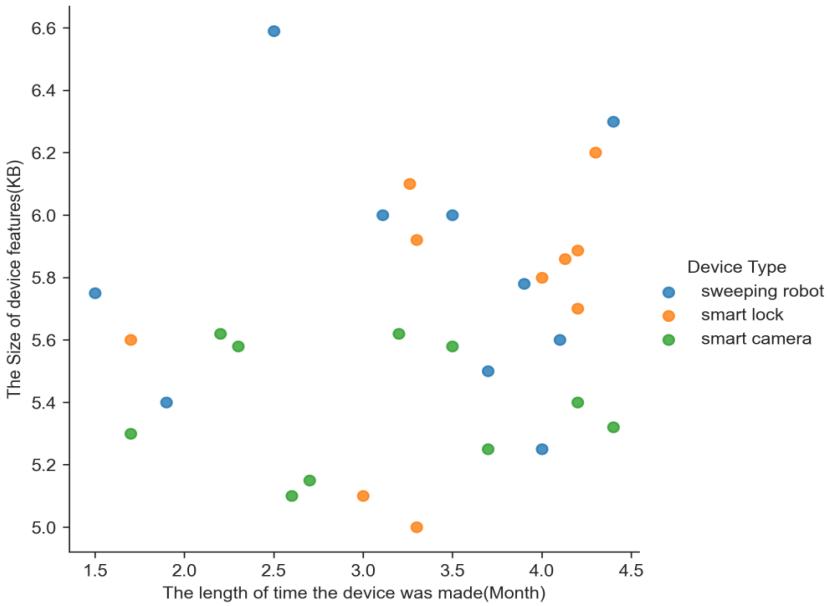

Fig. 5. The time when the devices are shipped and the size of their extracted features

We used a 32GB-RAM machine to deploy a blockchain platform with a throughput

(TPS) of more than 10000 to build the IoT Blockchain in our solution. In the device ID generation stage, the extracted information is generated by the SHA256 algorithm. The RSA algorithm is used to generate corresponding key pairs for vendors and users in the experiment. The corresponding role uses its private key to sign the digest generated by the hash algorithm and uploads the corresponding information to the Blockchain. They interact with Blockchain through the console program (simulation authentication platform). Besides, we use the SMS service to send the user a message verification code. Finally, we measure some data during the experiment under a stable network.

B. Experimental Results

In Fig. 6, we measured the time required for three types of edge devices features extraction, IoT device ID generation, and IoT device information registration. A total of 30 devices in three categories were measured. The time taken for features extraction of a single device was between 4000MS and 5400MS, and the time required for ID generation of each device was between 2500MS and 3700MS. The registration of each device information costs a minimum of 2600MS and a maximum of 3800MS. In the case of ignoring the network delay, it can be seen from the experimental results that the time required for the device information registration phase can be tolerated. The feature extraction and ID generation of the device are related to the feature size of the devices. The edge device information registration needs to interact with the blockchain and is related to the time of blockchain transaction confirmation.

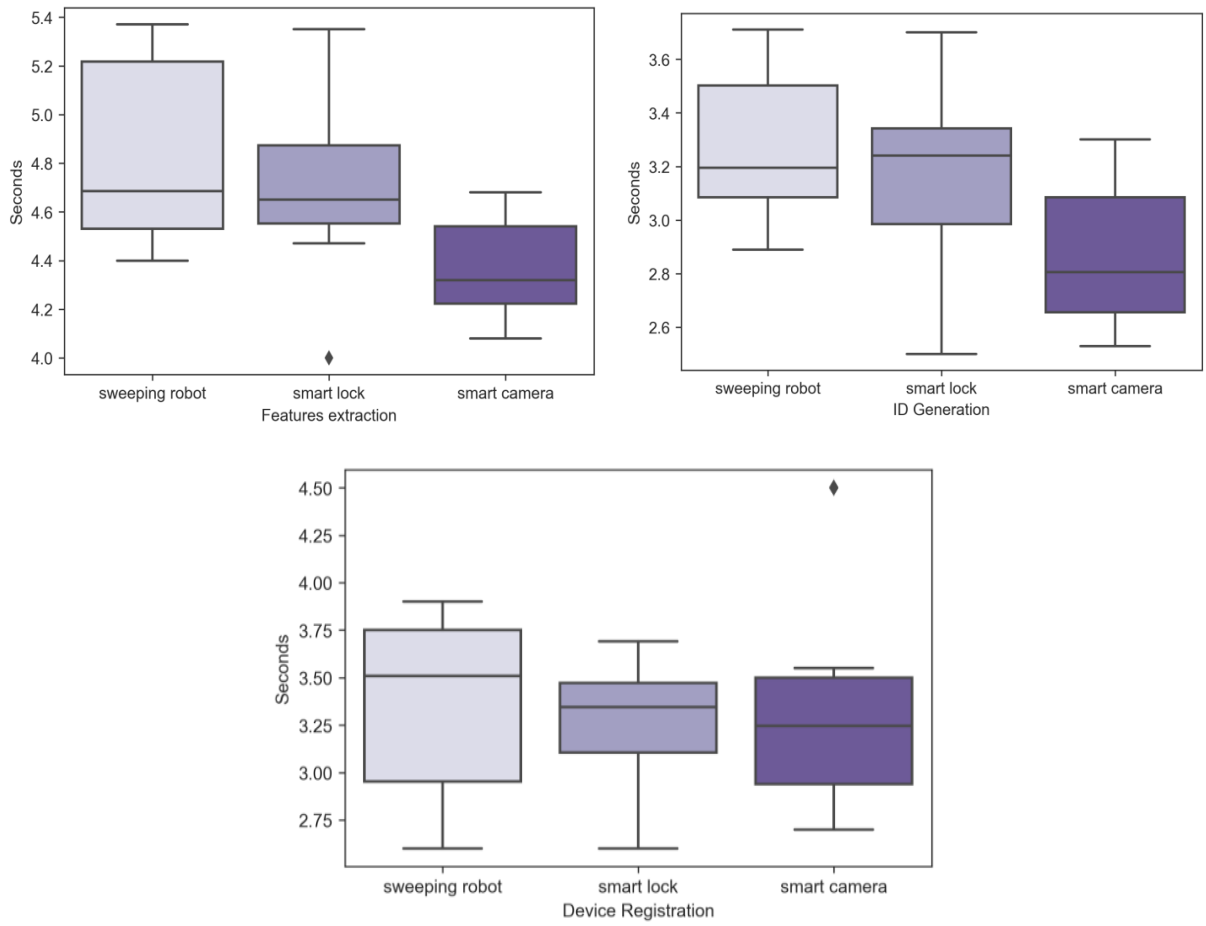

Fig. 6. The time consumption in IoT device registration steps

In the device authentication phase, we mainly measured the time it takes for different types of devices to be certified. In this process, time consumption is mainly in the process of querying and confirming device information. The result of our measurement is presented in Fig. 7. It can be seen that with the increase of devices, the overall time consumption also increases. And the average authentication time range of the equipment is between 2000MS and 3200MS.

Also, we compare the time consumption of user ID authentication based on blockchain and traditional account-password authentication in Fig. 8. It can be seen that as the number of connected devices increases, the blockchain-based method is better than the traditional account-password method for the average authentication

time of the former is less. The blockchain-based method only needs to verify whether the device ID corresponding to the user ID is consistent, while the traditional account verification method requires a certain query after entering the user account password, which may cause some delays and service requests.

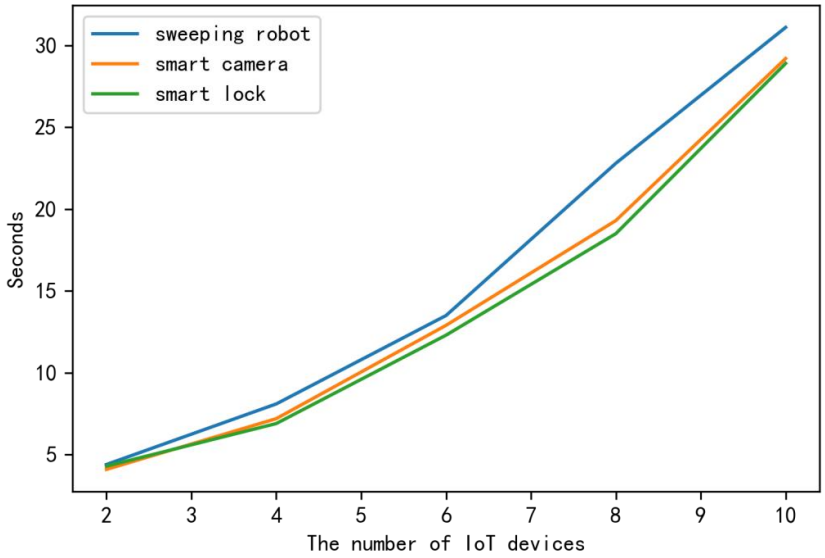

Fig. 7. Comparison of different types of IoT devices authentication

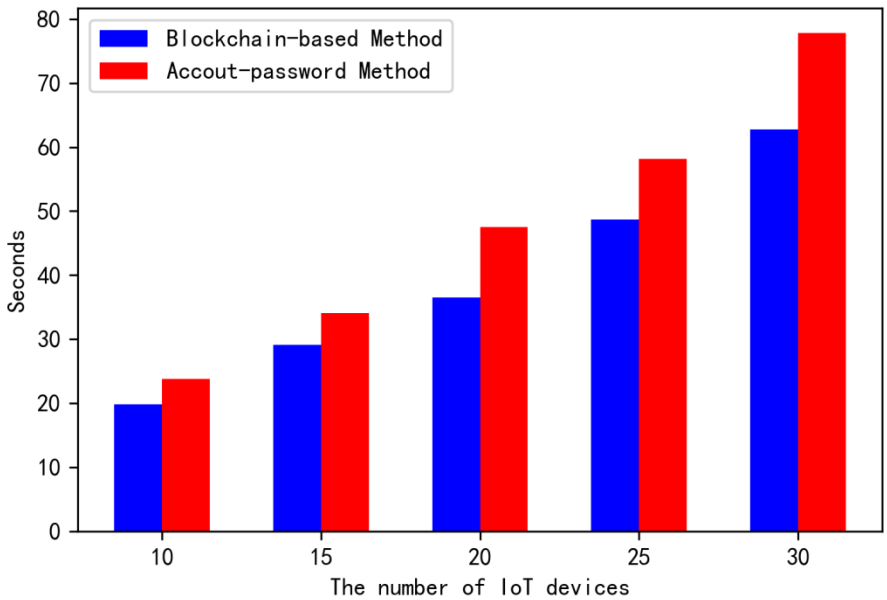

Fig. 8. Comparison of Blockchain-based method versus Account-password method

C. Performance analysis

(1) Security Analysis

It can be seen from the experiment that the proposed protocol can satisfy the authentication of multiple edge devices, and the required overhead is relatively low. Both the authentication of the edge devices and the users are completed within a few seconds, and most of the time is consumed by the actual setup of the device from the beginning, which leaves a small time window for potential attackers, thereby effectively protecting the user network from potential threats caused by vulnerable devices. Besides, our proposed solution is based on the blockchain infrastructure, which prevents the single-point server from being attacked by DDOS and other attacks. And only the hash of the relevant information is stored on the blockchain, which greatly protects the privacy of users. Finally, attackers can no longer use traditional brute force cracking to attack IoT devices. Even if they obtain other people's devices through improper means, they will eventually fail in the authentication. Because the information of the IoT device, such as the unique serial number, has been uploaded to the blockchain network by the IoT vendor before it leaves the factory, which provides an immutable certificate.

(2) Efficiency

In the proposed solution, the time cost required for the operation of IoT devices and user authentication is relatively low. On the other hand, compared to a large number of IoT devices based on account-password authentication methods, the proposed solution requires less resource consumption. Storing a large number of accounts and

passwords online requires massive resources. Various problems such as forgetting the password will hinder the user's authentication process too. Finally, the authentication of IoT devices from different vendors can be realized through our solution. Only one user ID is needed to realize the interconnection and intercommunication between different devices, avoiding repeated authentication of devices from multiple different vendors, and reducing the complexity of operation to improve the users' experience and the intelligence of the devices.

VII. CONCLUSIONS

This paper proposes a blockchain-based identity authentication protocol for edge IoT devices. In the proposed protocol, we use encryption and signatures based on the multi-dimensional features of edge devices to generate the device ID and then register the device information in the blockchain. When using the edge device for the first time, the user just needs to send an authentication request to the blockchain to verify whether the device is a trusted device. And we use the user ID to bind the edge IoT device to achieve efficient identity authentication of multiple IoT devices from different vendors based on the blockchain network. A series of experiments and the performance analysis have demonstrated the security and efficiency of our proposed protocol. By studying the merits of the proposed protocol, a series of tasks such as sustainable environment construction, smart identity authentication, authorization management among heterogeneous edge devices, and data sharing of different vendors and different types of edge IoT devices can be achieved.

ACKNOWLEDGMENT

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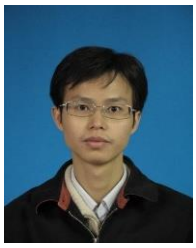

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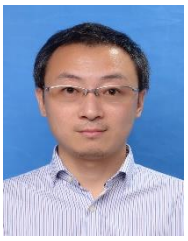

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